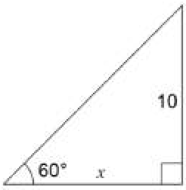
Trigonometry
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Find the exact value of x. 1



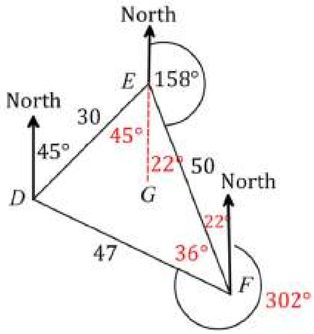
$\tan 60^\circ = \frac{10}{x}$

$\sqrt{3} = \frac{10}{x}$ $\therefore x = \frac{10}{\sqrt{3}}$ or $\frac{10\sqrt{3}}{3}$

2 Find the exact value of $\frac{\sin^2 60^o}{\cot 60^o + \sec 30^o}$. 2

$\frac{\frac{3}{4}}{\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}}$ — ① $\frac{\sqrt{3}}{4}$ — ①

3 Three girls are standing in the park. D is 30 metres from E and E is 50 metres from F. The bearing of E from D is 045° and the bearing of F from E is 158°.



a. Show all information given above on the diagram. 1

b. Show that $\angle DEF = 67°$. 1

$\angle DEG = 45°$

(Alternate angle to the bearing of E from D)

$\angle FEG = 180 - 158 = 22°$

(Straight angle of 180°)

$\angle DEF = \angle DEG + \angle FEG = 45 + 22 = 67°$

c. Show that the distance of F from D is approximately 47 metres. 1

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$DF^2 = 30^2 + 50^2 - 2 \times 30 \times 50 \times \cos 67 = 2227.8066$$
$$DF = 47.1996\ldots = 47 \text{ m}$$

$\therefore$ Distance of F from D is approximately 47 m

d. Hence, or otherwise, find the bearing of D from F. Answer to the nearest degree. 3

To find $\sin \angle DFE$

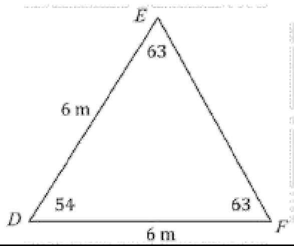
$$\frac{\sin \angle DFE}{30} = \frac{\sin 67}{47.1996\ldots}$$
$$\sin \angle DFE = \frac{\sin 67 \times 30}{47.1996\ldots}$$
$$\angle DFE = 35.8080\ldots \approx 36°$$

Bearing $= 360 - (36 + 22) = 302°$

(See above diagram)

$\therefore$ Bearing of D from F is 302°

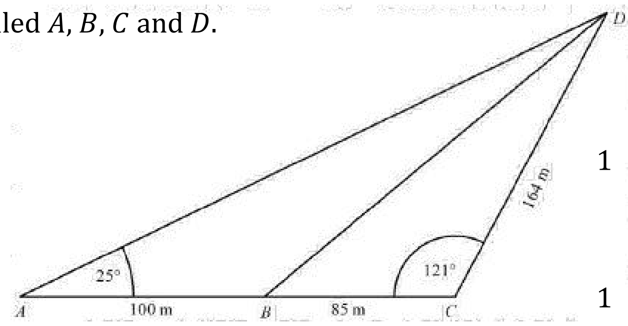
4 What is the area of the ΔDEF? Answer correct to two decimal places. 1



$$A = \frac{1}{2}ab\sin C$$
$$= \frac{1}{2} \times 6 \times 6 \times \sin 54° \approx 14.56 \text{ m}^2$$

5 The diagram below shows four cafés in a park, labelled A, B, C and D.

There are straight paths joining the cafés.



a. Show by calculations that the distance BD is 220 m (to the nearest metre). 1

b. Find the size of $\angle ADB$, correct to the nearest degree. 1

c. Lisa needs to walk from A to D. How many extra metres would she walk if she needs to go through B and C, rather than walking directly from A to D. 2

a)

$$BD^2 = 85^2 + 164^2 - 2 \times 85 \times 164 \times \cos 121°$$
$$= 48480.2615$$
$$BD = 220.182337 = 220 \text{ m (nearest metre)}$$

b)

$$\frac{\sin \angle ADB}{100} = \frac{\sin 25°}{220}$$
$$\sin \angle ADB = \frac{100 \sin 25°}{220}$$
$$= 0.1920992 \quad (0.19194 \text{ if using unrounded value })$$
$$\angle ADB = 11.07531 \quad (11.066) = 11°$$

c) Distance via B and $C = 100 + 85 + 164 = 349$ m

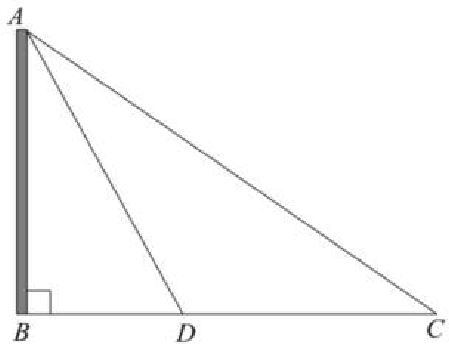
Compare with distance AD

Find AD using sine rule

$$\angle ABC = 180 - 25 - 11$$
$$= 144°$$
$$\frac{AD}{\sin 144°} = \frac{100}{\sin 11°}$$
$$AD = \frac{100 \sin 144°}{\sin 11°}$$
$$= 308.0490$$
$$= 308 \text{ m}$$

Extra distance $= 349 - 308 = \mathbf{41\ m}$

6 A vertical tower AB with points B, C and D in a straight line on the ground is shown below. The distance from CD is 100 metres. The angle of elevation to the top of the tower from point C is 35^o and from point D is 60^o.



a. Show that $AD = \frac{100sin35^o}{sin25^o}$. 2

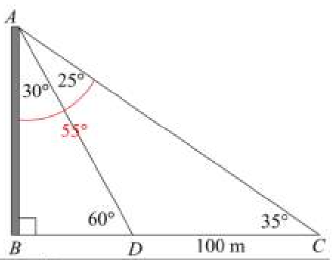
b. Calculate the height of the tower. Answer to the nearest metre. 2

(a)

$$\angle CAD = 55 - 30$$
$$= 25°$$

Use the sine rule in ΔADC

$$\frac{AD}{\sin 35} = \frac{100}{\sin 25}$$
$$AD = \frac{100 \sin 35}{\sin 25}$$

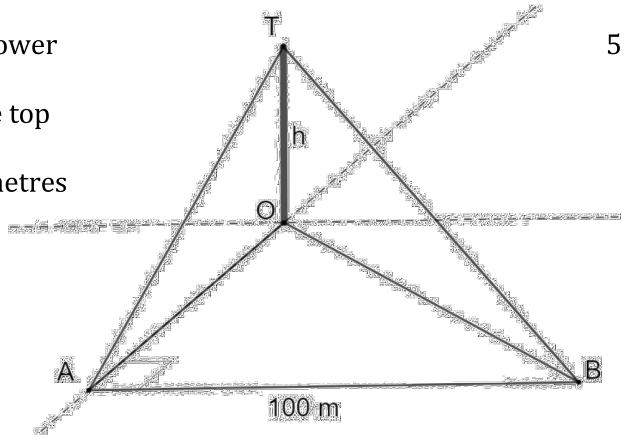


(b)

$$\sin 60 = \frac{AB}{AD}$$
$$AB = \frac{100 \sin 35 \sin 60}{\sin 25}$$
$$= 117.5367\ldots$$
$$\approx 118 \text{ m}$$

$\therefore$ The height of the tower is 118 metres.

7 Roger stands at a point A, which is due south of a tower OT of height h metres. The angle of elevation of the top of the tower from A is 45°. Roger then walks 100 metres due east to point B, from where he measures the angle of elevation of the top of the tower to be 30°. Calculate the bearing of B from the base of the tower at O. Give your answer correct to the nearest degree. 5



$$\frac{OT}{OA} = \tan 45° \Rightarrow OA = \frac{h}{\tan 45°} = h$$

$$\frac{OT}{OB} = \tan 30° \Rightarrow OB = \frac{h}{\tan 30°} = \sqrt{3}h$$

$$OB^2 = OA^2 + AB^2 \Rightarrow 3h^2 = h^2 + 100^2 \Rightarrow h = 50\sqrt{2}$$

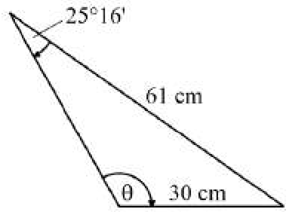
In right triangle OAB:

$$\tan \angle AOB = \frac{AB}{OA} = \frac{100}{50\sqrt{2}} = \sqrt{2}$$

$$\tan^{-1}\sqrt{2} = 55° \text{ (0 dp)}$$

Bearing of B from O: $180° - 55° = 125°$

8 In the following triangle, θ is an obtuse angle. Find the value of θ to the nearest minute. 3



Using sine rule:

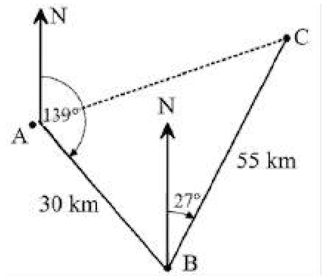
$$\frac{\sin sin\ \theta}{61} = \frac{\sin sin\ 25°16'}{30}$$

$$\sin sin\ \theta = 61 \times \frac{\sin sin\ 25°16'}{30} = 0.8678\ldots$$

$$\theta = 0.8678\ldots \approx 60°13'$$

As θ is obtuse, we take $180° - 60°13' = 119°47'$

9 A boat sails 30 kilometres from Island A to Island B on a bearing of 139°. It then sails 55 kilometres to Isla nd C on a bearing of 27° from Island B.



a. What is the size of $\angle ABC$? 1

$$\angle ABN = 180° - 139° = 41° \text{ (co-interior angles)}$$

b. Find the distance and bearing the boat must sail on to return to Island A from Island C. 3

$$AC^2 = 30^2 + 55^2 - 2\times30\times55\times \cos cos\ 68° = 2688.79\ldots$$

$$AC = \sqrt{2688.79\ldots} \approx 51.9\ km$$

Bearing:

Using sine rule to find $\angle ACB$:

$$\frac{\sin sin\ \theta}{30} = \frac{\sin sin\ 68°}{51.9}$$

$$\sin sin\ \theta = 30 \times \frac{\sin sin\ 68°}{51.9} = 0.5359\ldots$$

$$\theta = 0.5359\ldots \approx 32.4°$$

$\angle BCN = 180° - 27° = 153°$ (cointerior)

$\angle ACN = 153° - 32.4° = 120.6°$

Bearing is $360° - 120.6° = 239.4°$

The boat must sail 51.9 kilometres on a bearing of 239.4°.
(answer should be within 0.1 km and 0.1 degrees, depending on rounding and use of exact values)

Trigonometry Topic Test 2 Solution

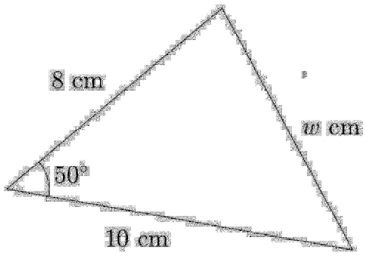
Total Marks: 30

Time: 50 minutes

1 Express 260^o as an exact radian value. 1

$260 \times \frac{\pi}{180} = \frac{13\pi}{9}$

2 The diagram shows a triangle with sides of length w cm, 8 cm and 10 cm along with an angle of 50^o. Use the cosine rule to calculate the value of w, correct to two significant figures. 2



$w^2 = 8^2 + 10^2 - 2 \times 8 \times 10 \times \cos 50$

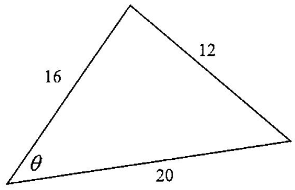
$= 64 + 100 - 160 \cos 50$

$= 7.8$ (2 s.g. fig.)

3 If $tan\theta = \frac{2}{3}$ and θ is acute, find the exact value of $cos\theta$. 1

$\frac{3}{\sqrt{13}}$

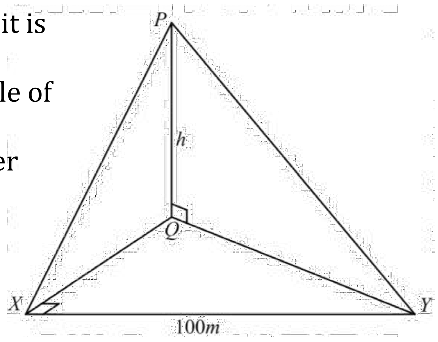
4 Find the value of θ, correct to the nearest minute. 2

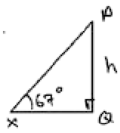


$\cos\theta = \frac{16^2 + 20^2 - 12^2}{2 \times 16 \times 20} = \frac{4}{5}$

$\theta = \cos^{-1}\left(\frac{4}{5}\right) = 36^o 52'$

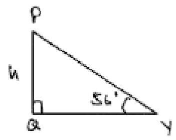
5 The angle of elevation of tower, PQ, from a point X due south of it is observed to be 67°. From another point Y, due east of X, the angle of elevation is 56°. Given $XY = 100\ m$, find the height h of the tower PQ to the nearest metre. 3





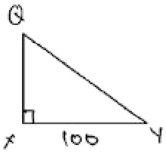
$\tan 67 = \frac{h}{XQ}$

$XQ = \frac{h}{\tan 67}$



$\tan 56 = \frac{h}{QY}$

$QY = \frac{h}{\tan 56°}$



$QX^2 + XY^2 = QY^2$

$\frac{h^2}{\tan^2 67} + 100^2 = \frac{h^2}{\tan^2 56}$

$\frac{h^2}{\tan^2 56} - \frac{h^2}{\tan^2 67} = 100^2$

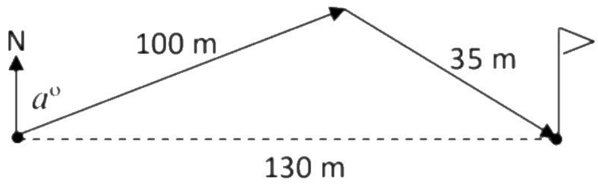
$h^2\left(\frac{1}{\tan^2 56} - \frac{1}{\tan^2 67}\right) = 100^2$

$h^2 = \frac{100^2}{\frac{1}{\tan^2 56} - \frac{1}{\tan^2 67}}$

$\therefore h = \sqrt{\frac{100^2}{\frac{1}{\tan^2 56} - \frac{1}{\tan^2 67}}}$

$\therefore h = 190.76 = \boxed{191\ m}$

6 On a gold course, the tee is a distance of 130 meters due west from the hole. On her first shot, Kelly hits the ball 100 meters but not at the correct angle. On her second shot she hits the ball 35 metres and gets it in the hole. On what bearing, α^o, did she hit her first shot? Give your answer correct to the nearest degree. 3



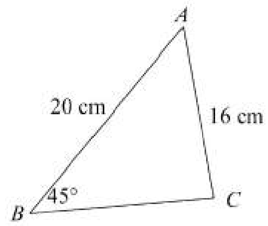
$\cos \theta = \dfrac{100^2 + 130^2 - 35^2}{2 \times 100 \times 130}$

$\theta \doteqdot 9°4'$

$\therefore a = 90° - 9°4'$

$\doteqdot 81°$ (nearest degree)

7 In triangle ABC, $AB = 20$ cm, $AC = 16$ cm and $\angle B = 45^o$. Find $\angle C$, correct to the nearest degree. 2



$\dfrac{\sin C}{20} = \dfrac{\sin 45}{16}$

$\sin C = \dfrac{\sqrt{2}}{32} \times 20$

$= \dfrac{5\sqrt{2}}{8}$

$\therefore \angle C = \sin^{-1} \dfrac{5\sqrt{2}}{8}$

$= 62°7'$ or $180° - 62°7' = 117°53'$

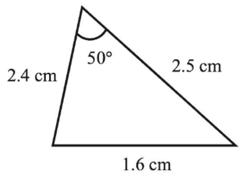
$= 62°$ or $118°$ (nearest degree)

8 What is the value of $tanx$ given that $sinx = \frac{1}{3}$ for $\frac{\pi}{2} < x < \pi$? 1

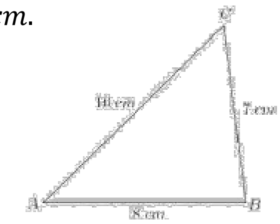
$-\dfrac{1}{2\sqrt{2}}$

9 Find the area of the triangle. 1

$$\frac{1}{2} \times 2.4 \times 2.5 \times \sin 50°$$



10 The diagram shows ΔABC with sides $AB = 8\ cm$, $BC = 7\ cm$ and $AC = 10\ cm$.



a. Show that $cosA = \frac{23}{32}$. 1

$$\cos A = \frac{8^2 + 10^2 - 7^2}{2 \times 8 \times 10} = \frac{23}{32}$$

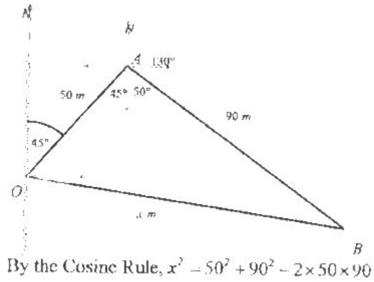
b. By finding the exact value of $sinA$, determine the exact value of the area of ΔABC. 2

$\sin A = \dfrac{\sqrt{32^2 - 23^2}}{32}$ $\qquad Area = \dfrac{1}{2} \times 8 \times 10 \times \sin A$

$\sin A = \dfrac{\sqrt{495}}{32}$ $\qquad Area = \dfrac{1}{2} \times 8 \times 10 \times \dfrac{3\sqrt{55}}{32}$

$\sin A = \dfrac{3\sqrt{55}}{32}$ $\qquad Area = \dfrac{15\sqrt{55}}{4}\ units^2$

11 A drone is used during the filming of a TV show. The drone leaves its base station, at point O, and flies 50 m on a bearing of $045^{o}T$ to point A. It then changes direction to $130^{o}T$ and flies a further 90 m to point B. To the nearst metre, calculate how far the drone is from base when it reaches point B. 3

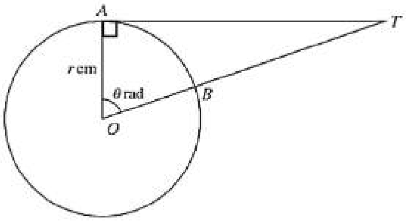


By the Cosine Rule, $x^2 = 50^2 + 90^2 - 2\times 50\times 90\cos(45^\circ + 50^\circ)$

$= 11384.40168...$

$\therefore x = 106.6977$, i.e. $x = 107$ m, correct to nearest metre

12 The diagram shows a circle with centre O and radius r cm. The points A and B lie on the circle and AT is a tangent to the circle. OBT is a straight line and $\angle AOB = \theta$ radians.



a. Express the area of the shaded region in terms of r and θ. 2

$\tan\theta = \frac{AT}{r}$ $\therefore AT = r\tan\theta$

Shaded area $=$ area$_{\triangle OAT}$ $-$ sector$_{OAT}$

$= \frac{1}{2}\times r\times r\tan\theta - \frac{1}{2}r^2\theta$

$= \frac{1}{2}r^2\tan\theta - \frac{1}{2}r^2\theta$

b. Given that $r = 3$ and $\theta = 1.2$, find the perimeter of the shaded region. Give your answer to 2 decimal places. 2

Perimeter $= AT + BT +$ arc AB

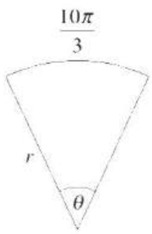
$= 3\tan 1.2 + (OT - OB) + 3\times 1.2$

$= 3\tan 1.2 + \left(\frac{3}{\cos 1.2} - 3\right) + 3\times 1.2$ $\quad \cos\theta = \frac{r}{OT}$

$= 16.5955....$ $\quad OT = \frac{3}{\cos 1.2}$

$= 16.60$ cm

13 The diagram shows a sector with radius r and angle θ where $0 < \theta \leq 2\pi$. The arc length is $\frac{10\pi}{3}$.



a. Find the value of θ and hence, show that $r \geq \frac{5}{3}$. 2

$l = r\theta$ where $0 < \theta \leq 2\pi$

$\therefore \frac{10\pi}{3} = r\theta$

$\theta = \frac{10\pi}{3r}$

Now $0 < \theta \leq 2\pi$

$0 < \frac{10\pi}{3r} \leq 2\pi$

$0 < 10\pi \leq (2\pi\times 3r)$

$0 < \frac{10\pi}{2\pi} \leq 3r$

$0 < \frac{5}{3} \leq r$

$\therefore r \geq \frac{5}{3}$

b. Calculate the ar

$A = \frac{1}{2}r^2\theta$ where $\theta = \frac{10\pi}{3r}$ and $r = 4$

$A = \frac{1}{2}\times 4^2 \times \frac{10\pi}{...} = \frac{1}{2}\times 16\times \frac{10\pi}{...}$